

MAKE YOUR STAR WHEEL

A SKY CALCULATOR FOR STATE COLLEGE, PA

YOU WILL NEED

- Pieces 1 and 2, cut from the back of this book or printed from the code.
- Thin cardboard, like a cereal box, and a glue stick (or print the pieces on cardstock).
- Scissors, a pencil, and one brass paper fastener.
- A red flashlight for reading it outside.

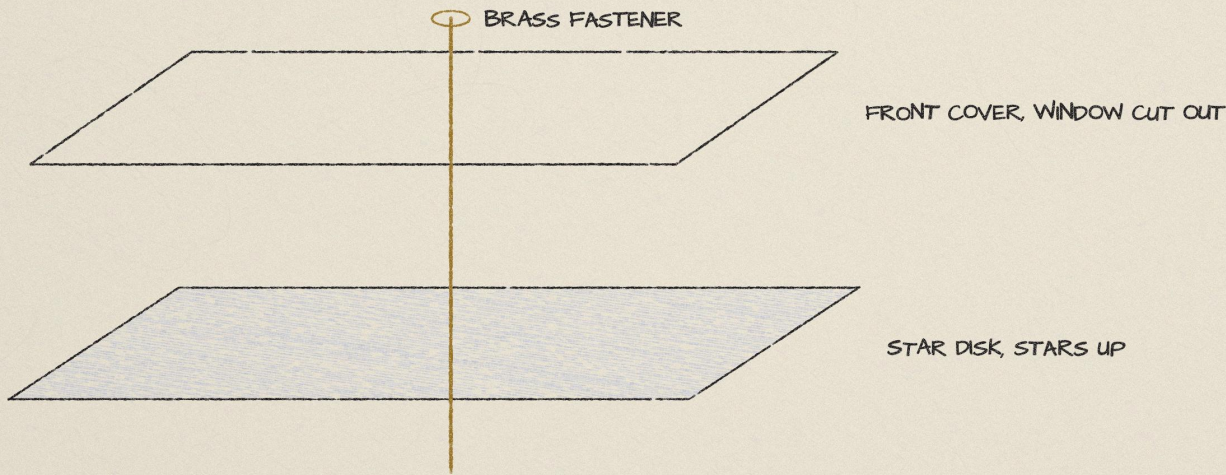


Scan to print
Pieces 1 and 2.
[treyjg/flat-earth](https://github.com/treyjg/flat-earth)

MAKE IT

- 1 Glue each piece onto its own piece of thin cardboard and let them dry flat. Printed on cardstock? Skip this step.
- 2 Cut out the star disk along its dashed circle.
- 3 Cut out the front cover. Then cut out its window, but keep the thin bridge and the round pad in the middle: they hold the fastener.
- 4 Poke a small hole through the red cross on each piece with a sharp pencil.
- 5 Stack the cover on top of the disk, stars facing up. Push the fastener through both holes and open its legs on the back. Leave it loose enough to turn.

HOW THE PIECES STACK



Tip: the thin bridge holds the round pad, but it also hides a strip of sky. If you have clear plastic, like a page protector, tape it over the whole window. The plastic holds the pad in place, so you can cut the bridge away and see the whole sky.

TRY IT

- | | |
|-----------------------------|--|
| January 15, 9 PM | Orion stands high in the south, with Sirius below and to the left. |
| April 15, 10 PM (red time) | The Big Dipper is nearly overhead; follow its handle to Arcturus. |
| July 15, 10 PM (red time) | Vega is almost straight up, the corner of the Summer Triangle. |
| October 15, 9 PM (red time) | The Great Square of Pegasus is high in the south-east. |

USE IT

- 1 Turn the disk until today's date, on the rim, lines up with the time on the cover. Use the black times in winter and the red times during daylight saving time.
- 2 Face a direction: north, south, east or west. Hold the wheel over your head with that word at the bottom, nearest you.
- 3 The window now shows the whole sky. Its edge is the horizon, and the stars near the bottom edge are the ones low in front of you.
- 4 The middle of the window, a little below the bridge, is the spot straight overhead.

GOOD TO KNOW

- It is made for 41 degrees north, so it works well from Virginia to New York, including Cherry Springs State Park.
- State College is west of the middle of the Eastern time zone, so its sky runs about 11 minutes behind the clock. This wheel has that built in.
- Planets and the Moon are not on the disk because they wander. If you see a bright "star" near the red dashed line, it is probably a planet.

Print at actual size. Both pieces must be printed at the same size.

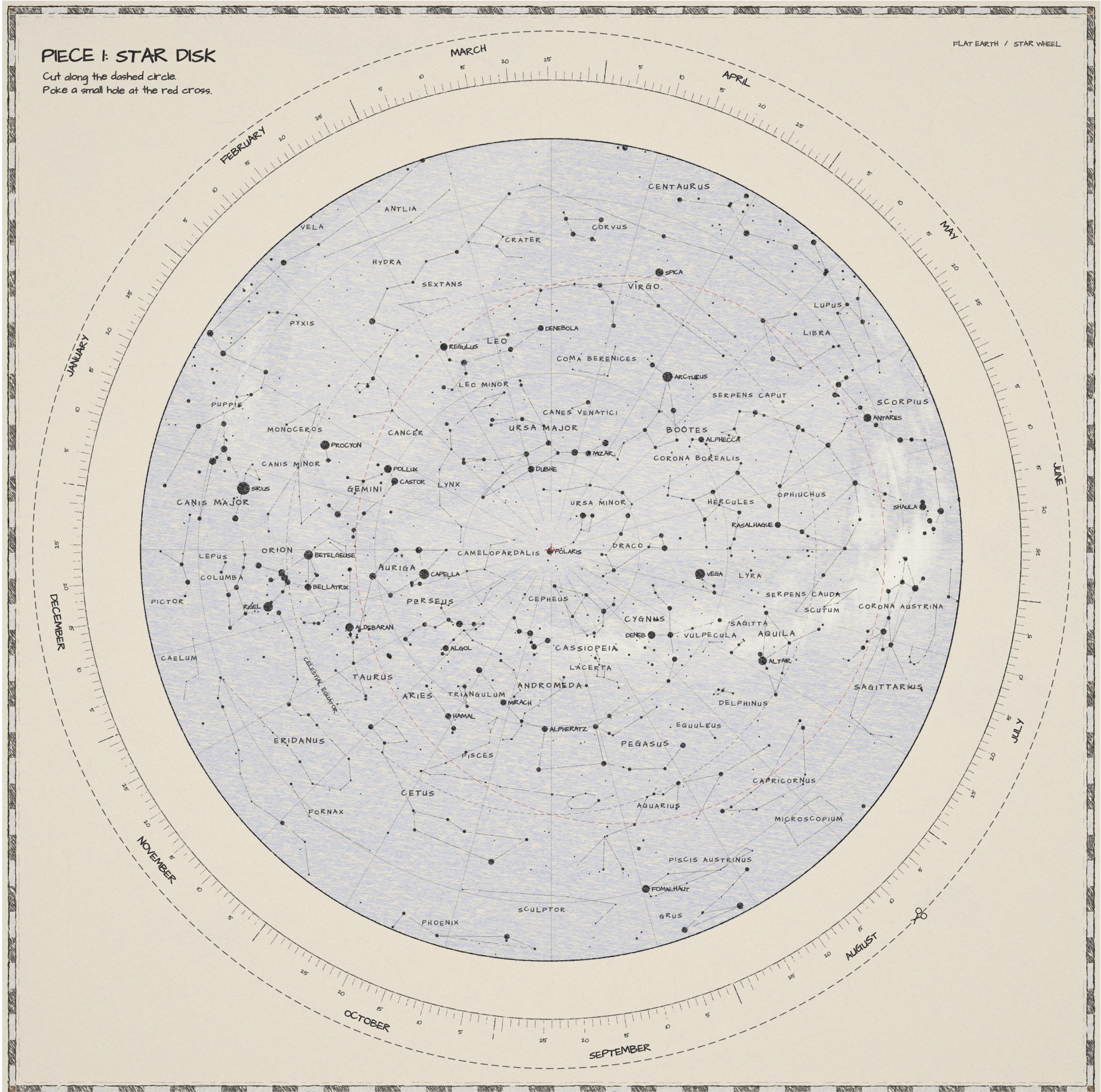


This bar should measure 2 inches. If it does not, your printer changed the size.

PIECE 1: STAR DISK

Cut along the dashed circle.
Poke a small hole at the red cross.

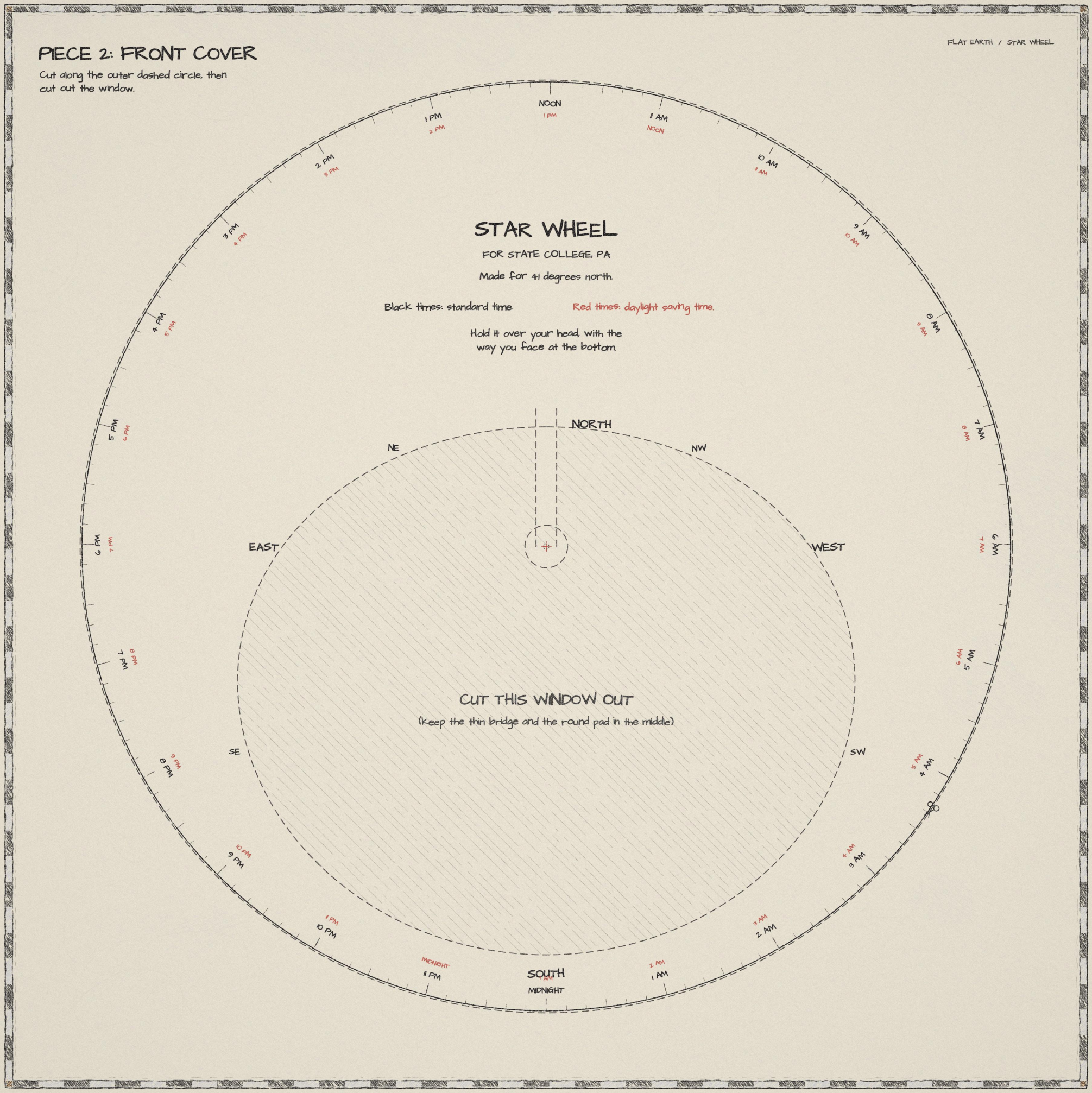
FLAT EARTH / STAR WHEEL



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